

MA3632 Lecture 12 – Graph Analytics

Every dataset studied so far in this module has been represented as a matrix of feature vectors: rows are observations, columns are measured quantities, and the relationships between observations are inferred indirectly, through distance or correlation. Many real datasets, however, come with relationships already specified explicitly: who follows whom on a social network, which proteins interact, which web pages link to which, which customers purchase the same products. A *graph* represents these relationships directly as the primary object of study, rather than as a derived quantity. This lecture is self-contained and assumes no prior exposure to graph theory: it is exactly the material promised in Lecture 1, where graph-analytic methods were named explicitly as being “introduced later in the module.”

The lecture builds in three stages. First, graphs are defined as mathematical objects, along with the structural notions – degree, paths, connectivity – needed to reason about them. Second, we ask which vertices of a graph are “important,” leading to centrality measures and to PageRank. Third, we ask how a graph can be partitioned into clusters, which leads to the graph Laplacian and spectral clustering. This last topic is the deliberate capstone of the module: it reuses the eigendecomposition machinery of Lecture 4 and combines it with the k -means clustering objective, applied to a genuinely new object.

Remark 0.1. Unless otherwise stated, the definitions, propositions, and theorems in Sections 1–3 concern **simple, unweighted** graphs, for which $\deg(v) = \sum_j A_{vj}$ is literally the number of edges incident to v . Weighted graphs are introduced alongside the unweighted case below because they arise naturally in applications, but for a weighted graph $\sum_j A_{vj}$ is the *weighted degree* (or *strength*) of v – a sum of edge weights, not a count of edges – and several results stated for unweighted graphs (in particular Proposition 2.2 on counting walks) do not carry over verbatim to the weighted case. We flag explicitly where a result depends on the unweighted assumption.

1 Graphs as Mathematical Objects

Definition 1.1. A **graph** $G = (V, E)$ consists of a finite set of **vertices** V and a set of **edges** E , where each edge is a pair of vertices. In an **undirected** graph, an edge $\{i, j\}$ is unordered, so $\{i, j\} = \{j, i\}$. In a **directed** graph (or *digraph*), an edge (i, j) is ordered, representing a connection from i to j that need not be reciprocated. A graph is **weighted** if each edge carries a non-negative real number $w_{ij} \geq 0$ (representing, e.g., strength of interaction or inverse physical distance); otherwise it is **unweighted**, which we treat as the special case $w_{ij} \in \{0, 1\}$. (Signed networks, in which weights may be negative, require separate treatment and are not considered in this lecture.)

Definition 1.2. The **adjacency matrix** of a graph G with $n = |V|$ vertices is the $n \times n$ matrix A with

$$A_{ij} = \begin{cases} w_{ij} & \text{if } (i, j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

For an undirected graph, A is symmetric, since $\{i, j\} \in E$ implies both A_{ij} and A_{ji} equal w_{ij} . For a directed graph, A need not be symmetric.

Definition 1.3. For a **simple unweighted** undirected graph, the **degree** of a vertex v , $\deg(v)$, is the number of edges incident to v , equivalently $\deg(v) = \sum_j A_{vj}$. In a directed unweighted graph,

the **out-degree** of v is $\sum_j A_{vj}$ and the **in-degree** is $\sum_i A_{iv}$. For a weighted graph, the same sums $\sum_j A_{vj}$ (and, for digraphs, $\sum_i A_{iv}$) define the **weighted degree** or **strength** of v , a sum of edge weights rather than a count of edges; the two notions coincide exactly when all weights are 0 or 1.

Proposition 1.4 (Handshake lemma). *For a simple unweighted undirected graph, $\sum_{v \in V} \deg(v) = 2|E|$.*

Proof. Each edge $\{i, j\} \in E$ contributes exactly 1 to $\deg(i)$ and exactly 1 to $\deg(j)$, so it contributes exactly 2 to the sum $\sum_v \deg(v)$. Summing this contribution over all $|E|$ edges gives $\sum_v \deg(v) = 2|E|$. $\square$

Example 1.5. Let $V = \{1, 2, 3, 4\}$ and $E = \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{3, 4\}\}$. The degree sequence is $\deg(1) = 2$, $\deg(2) = 2$, $\deg(3) = 3$, $\deg(4) = 1$, summing to $8 = 2 \times 4$, consistent with Proposition 1.4 since $|E| = 4$. The adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

2 Walks, Connectivity, and Components

Definition 2.1. A **walk** of length k from vertex i to vertex j is a sequence of vertices $i = v_0, v_1, \dots, v_k = j$ such that $(v_{t-1}, v_t) \in E$ for each $t = 1, \dots, k$. A walk in which no vertex is repeated is a **path**.

Proposition 2.2. *For a simple unweighted graph with adjacency matrix A , the entry $(A^k)_{ij}$ equals the number of walks of length k from vertex i to vertex j . (For a weighted graph, $(A^k)_{ij}$ is instead the sum, over all such walks, of the product of the edge weights along each walk.)*

Proof. By induction on k . For $k = 1$, $(A^1)_{ij} = A_{ij}$, which is exactly the number of edges from i to j (0 or 1 for a simple unweighted graph), i.e. the number of walks of length 1. Assume the claim holds for k . A walk of length $k + 1$ from i to j consists of a walk of length k from i to some intermediate vertex m , followed by an edge from m to j . By the inductive hypothesis, the number of length- k walks from i to m is $(A^k)_{im}$, and the number of edges from m to j is A_{mj} . Summing over all possible intermediate vertices m and using matrix multiplication,

$$(A^{k+1})_{ij} = \sum_{m=1}^n (A^k)_{im} A_{mj},$$

which counts every length- $(k + 1)$ walk from i to j exactly once, grouped by its penultimate vertex m . $\square$

Definition 2.3. Two vertices i and j are **connected** if there exists a walk from i to j (equivalently, since the empty walk connects any vertex to itself and undirected edges are traversable in either direction, if $i = j$ or a path exists between them). A graph is **connected** if every pair of vertices is connected.

Proposition 2.4. *Connectivity is an equivalence relation on V , and its equivalence classes are called the **connected components** of G .*

Proof. Reflexivity holds because the length-0 walk connects any vertex to itself. Symmetry holds for undirected graphs because reversing the vertex sequence of a walk from i to j gives a walk from j to i . Transitivity holds because a walk from i to m followed by a walk from m to j is a walk from i to j . An equivalence relation partitions V into disjoint equivalence classes, which are precisely the maximal subsets of mutually connected vertices, i.e. the connected components. $\square$

Remark 2.5. Connected components can be computed constructively by breadth-first or depth-first search from an arbitrary starting vertex, visiting every vertex reachable from it before moving to an unvisited vertex and repeating. This is how `networkx`'s `connected_components` function operates in practice; the equivalence-relation argument above is what guarantees the resulting partition is well defined regardless of which vertex the search starts from.

3 Centrality Measures

Centrality measures quantify how “important” a vertex is within a graph. Different measures formalise different notions of importance, and a vertex ranked highly by one measure need not be ranked highly by another.

3.1 Degree centrality

Definition 3.1. The **degree centrality** of vertex v is $\deg(v)/(n-1)$, the vertex's degree normalised by the maximum possible degree in a simple graph on n vertices.

Degree centrality is the simplest and cheapest measure to compute, but it only captures local structure: it cannot distinguish a vertex connected to three peripheral vertices from a vertex connected to three highly connected hubs.

3.2 Closeness centrality

Definition 3.2. Let $d(v, u)$ denote the length of the shortest path between vertices v and u in a connected graph. The **closeness centrality** of v is

$$C(v) = \frac{n-1}{\sum_{u \neq v} d(v, u)}.$$

Closeness centrality is large when a vertex is, on average, only a few steps from every other vertex. It requires the graph to be connected (or is applied component-wise), since $d(v, u)$ is undefined between vertices in different components.

3.3 Betweenness centrality

Definition 3.3. For vertices $s \neq t$, let σ_{st} denote the number of shortest paths between s and t , and let $\sigma_{st}(v)$ denote the number of those shortest paths that pass through vertex v (with $v \neq s, t$). The **betweenness centrality** of v is

$$B(v) = \sum_{s \neq t, s, t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}}.$$

Betweenness centrality is large for vertices that lie on many shortest paths between other pairs of vertices – vertices that act as bridges or bottlenecks in the network, even if their own degree is

modest. Betweenness centrality in this form is due to Freeman (1977); Freeman (1978) gives the companion conceptual clarification unifying degree, closeness, and betweenness centrality as a single family of formal, comparable measures.

3.4 Eigenvector centrality

Degree centrality treats all neighbours as equally valuable. Eigenvector centrality instead proposes that a vertex is important if it is connected to other important vertices, giving a self-referential definition that has to be solved rather than merely evaluated.

Definition 3.4. The **eigenvector centrality** of a connected undirected graph with adjacency matrix A is the vector x satisfying $Ax = \lambda x$ for the largest eigenvalue λ of A , normalised so that x has unit norm and non-negative entries.

Since A is real and symmetric, the spectral theorem used throughout Lecture 4 guarantees A has n real eigenvalues and an orthonormal basis of real eigenvectors. What is not immediate from the spectral theorem alone is that the eigenvector belonging to the *largest* eigenvalue can be chosen with strictly positive entries and is, up to scalar multiplication, the unique eigenvector of A with this property – which is what makes Definition 3.4 well posed.

Theorem 3.5 (Perron–Frobenius). *If A is the adjacency matrix of a connected undirected graph, then its spectral radius $\rho(A)$ is a positive, simple eigenvalue of A and has an eigenvector with strictly positive entries, unique up to scalar multiplication. If the graph is additionally non-bipartite, no other eigenvalue of A has modulus $\rho(A)$.*

Theorem 3.5 is a special case of the general Perron–Frobenius theorem for non-negative irreducible matrices; connectivity of the graph is exactly the condition that makes A irreducible. We use the theorem as stated, without reproving it here, and take it as the justification for why Definition 3.4 picks out a unique, sign-consistent centrality vector. The non-bipartite qualification in the final sentence matters: a bipartite connected graph (a star, for instance) always has $-\rho(A)$ as an eigenvalue alongside $\rho(A)$, so the *largest* eigenvalue is still simple with a positive eigenvector, but it is not the *unique* eigenvalue of largest modulus in that case.

Example 3.6. Consider the star graph on 4 vertices, with centre c joined to three leaves ℓ_1, ℓ_2, ℓ_3 and no edges between leaves. By the symmetry between the three leaves, an eigenvector centrality solution has the form x_c for the centre and a common value x_ℓ for every leaf. The eigenvector equation $Ax = \lambda x$ gives, at the centre, $\lambda x_c = 3x_\ell$ (the centre’s three neighbours are the leaves), and at each leaf, $\lambda x_\ell = x_c$ (each leaf’s only neighbour is the centre). Substituting $x_c = \lambda x_\ell$ into the first equation gives $\lambda^2 x_\ell = 3x_\ell$, so $\lambda^2 = 3$ and, taking the positive root as required by Theorem 3.5, $\lambda_{\max} = \sqrt{3}$. Then $x_c = \sqrt{3}x_\ell$. Normalising $x_c^2 + 3x_\ell^2 = 1$ gives $3x_\ell^2 + 3x_\ell^2 = 1$, so $x_\ell = 1/\sqrt{6}$ and $x_c = 1/\sqrt{2}$. The centre’s eigenvector centrality ($1/\sqrt{2} \approx 0.707$) is more than twice each leaf’s ($1/\sqrt{6} \approx 0.408$), reflecting the intuition that the centre is the only vertex connected to anything of consequence. The star is bipartite (centre versus leaves), so $-\sqrt{3}$ is also an eigenvalue of A , alongside $\sqrt{3}$ and a repeated eigenvalue 0; this is the situation the non-bipartite qualification of Theorem 3.5 is guarding against, though it does not affect this example’s use of the positive eigenvector belonging to $\lambda_{\max} = \sqrt{3}$ itself.

Remark 3.7. In `networkx`, eigenvector centrality is computed by power iteration rather than direct eigendecomposition, since real graphs are typically too large to diagonalise exactly:

```
import networkx as nx
G = nx.star_graph(3) # centre 0, leaves 1, 2, 3
centrality = nx.eigenvector_centrality(G)
```

For the star graph of Example 3.6, this recovers the values computed by hand, up to the library’s own normalisation convention.

4 PageRank

Eigenvector centrality, as defined above, is designed for undirected graphs. On directed graphs, however, the eigenvector-centrality idea encounters additional complications: the adjacency matrix may be reducible, so Theorem 3.5 no longer applies and a positive leading eigenvector need not exist uniquely (or at all); and dangling nodes – vertices with no outgoing edges – create difficulties for a random-walk interpretation, since a walker reaching one has nowhere to go. PageRank, introduced by Brin and Page (1998) to rank web pages by the link structure of the web, repairs both problems by mixing the eigenvector-centrality idea with a small amount of uniform random restart.

Definition 4.1. Let A be the adjacency matrix of a directed graph in which $A_{ij} = 1$ if there is a link from j to i (note the reversed indexing convention relative to Definition 1.2, chosen so that column j describes where vertex j ’s links point). Let d_j be the out-degree of vertex j , and define the transition matrix M by

$$M_{ij} = \begin{cases} A_{ij}/d_j, & d_j > 0, \\ 1/n, & d_j = 0, \end{cases}$$

so that a vertex with no outgoing links (a *dangling node*) is treated as linking uniformly to every vertex, including itself; this makes every column of M sum to 1, i.e. M is column-stochastic. For a **damping factor** $\alpha \in (0, 1)$ (conventionally $\alpha = 0.85$), the **Google matrix** is

$$G = \alpha M + (1 - \alpha) \frac{1}{n} J,$$

where J is the $n \times n$ all-ones matrix. Since M and J/n are both column-stochastic, so is G : each column of G is a convex combination (α and $1 - \alpha$) of two columns that each sum to 1. The **PageRank vector** r is the stationary distribution satisfying $Gr = r$ with $r_i \geq 0$ and $\sum_i r_i = 1$.

Remark 4.2. The term $(1 - \alpha)J/n$ can be read as a “random surfer” who, at each step, follows an outgoing link with probability α and otherwise jumps to a uniformly random vertex in the graph; at a dangling node, with no outgoing links to follow, the dangling-node convention above has the surfer jump to a uniformly random vertex regardless, which is exactly what keeps every column of M – and hence of G – summing to 1. This teleportation step, together with the dangling-node correction, guarantees G is a strictly positive stochastic matrix, which by the Perron–Frobenius theorem for positive matrices has a unique stationary distribution regardless of the underlying graph’s connectivity – this is precisely what plain eigenvector centrality lacked.

Example 4.3. Let $V = \{A, B, C\}$ with directed edges $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Every vertex here has at least one outgoing edge, so the dangling-node correction plays no role in this example. Vertex A has out-degree 2, so $M_{BA} = M_{CA} = \frac{1}{2}$; vertex B has out-degree 1, so $M_{CB} = 1$; vertex C has out-degree 1, so $M_{AC} = 1$. Taking $\alpha = \frac{1}{2}$ for arithmetic convenience,

$$G = \frac{1}{2}M + \frac{1}{6}J = \begin{pmatrix} 1/6 & 1/6 & 2/3 \\ 5/12 & 1/6 & 1/6 \\ 5/12 & 2/3 & 1/6 \end{pmatrix}.$$

Solving $Gr = r$ together with $r_A + r_B + r_C = 1$ gives $r_A = 14/39$, $r_B = 10/39$, $r_C = 15/39$. Vertex C ranks highest because it receives incoming links from both A and B ; vertex A ranks above B despite both having in-degree 1, because A 's sole incoming link comes from C , which is itself highly ranked. This illustrates the recursive character PageRank shares with eigenvector centrality: rank flows from already well-ranked vertices.

Remark 4.4. `import networkx as nx`
`G = nx.DiGraph([('A','B'), ('A','C'), ('B','C'), ('C','A')])`
`pagerank = nx.pagerank(G, alpha=0.85)`

A conventional choice in PageRank implementations is $\alpha = 0.85$ rather than the $\alpha = \frac{1}{2}$ used in Example 4.3; a smaller α damps the influence of link structure more heavily and pulls the ranking closer to uniform.

5 The Graph Laplacian and Spectral Clustering

The final question this lecture addresses is how to partition a graph into clusters of densely connected vertices with comparatively few edges between clusters. The tool that answers this is a matrix built from the adjacency matrix and the degree sequence, whose eigendecomposition – the same basic linear-algebraic operation used to obtain principal components in Lecture 4 – reveals the graph's cluster structure directly, recovering the connected components of Proposition 2.4 exactly as a special case (Theorem 5.4 below).

Definition 5.1. Let D be the diagonal matrix with $D_{vv} = \deg(v)$. The (unnormalised) **graph Laplacian** is $L = D - A$.

Proposition 5.2. For any $x \in \mathbb{R}^n$,

$$x^\top Lx = \sum_{\{i,j\} \in E} (x_i - x_j)^2.$$

Consequently L is symmetric positive semi-definite, and $L\mathbf{1} = 0$, where $\mathbf{1}$ is the all-ones vector.

Proof. Expanding the right-hand side,

$$\sum_{\{i,j\} \in E} (x_i - x_j)^2 = \sum_{\{i,j\} \in E} (x_i^2 + x_j^2 - 2x_i x_j) = \sum_v \deg(v) x_v^2 - 2 \sum_{\{i,j\} \in E} x_i x_j,$$

since each vertex v contributes x_v^2 once for every edge incident to it, i.e. $\deg(v)$ times in total. The first term is $x^\top D x$. For the second term, note that $x^\top A x = \sum_{i,j} A_{ij} x_i x_j$ counts each edge $\{i, j\}$ twice in the double sum (once as (i, j) , once as (j, i)), so $x^\top A x = 2 \sum_{\{i,j\} \in E} x_i x_j$. Combining, $\sum_{\{i,j\} \in E} (x_i - x_j)^2 = x^\top D x - x^\top A x = x^\top L x$. Since a sum of squares is non-negative, $x^\top L x \geq 0$ for every x , so L is positive semi-definite. Taking $x = \mathbf{1}$ gives $x_i - x_j = 0$ for every edge, so $\mathbf{1}^\top L \mathbf{1} = 0$; since L is symmetric positive semi-definite this forces $L\mathbf{1} = 0$ (Proposition 5.3 below makes this implication precise). $\square$

Proposition 5.3. For a symmetric positive semi-definite matrix L , $x^\top L x = 0$ if and only if $Lx = 0$.

Proof. Write the spectral decomposition $L = \sum_{k=1}^n \lambda_k v_k v_k^\top$ with $\lambda_k \geq 0$ and $\{v_k\}$ an orthonormal eigenbasis (Lecture 4). Then $x^\top L x = \sum_k \lambda_k (v_k^\top x)^2$, a sum of non-negative terms. This sum is zero if and only if every term with $\lambda_k > 0$ has $v_k^\top x = 0$, i.e. x lies entirely in the span of eigenvectors with $\lambda_k = 0$. But that is exactly the condition $Lx = \sum_k \lambda_k (v_k^\top x) v_k = 0$. $\square$

Theorem 5.4. *The multiplicity of the eigenvalue 0 of L equals the number of connected components of G , and its eigenspace is spanned by the indicator vectors of the connected components.*

Proof. By Propositions 5.2 and 5.3, $Lx = 0$ if and only if $x_i = x_j$ for every edge $\{i, j\} \in E$. If i and j lie in the same connected component, there is a path between them, and applying $x_i = x_j$ along each edge of the path forces $x_i = x_j$; hence x is constant on each connected component. Conversely, any vector that is constant on each component satisfies $x_i = x_j$ for every edge, since every edge joins two vertices in the same component, so $Lx = 0$. The indicator vectors of the components are linearly independent (they have disjoint supports) and span exactly this set of constant-on-each-component vectors, so they form a basis for the null space of L , and the number of connected components equals $\dim(\ker L)$, the multiplicity of eigenvalue 0. $\square$

Theorem 5.4 is the exact statement for a graph that is already disconnected. Its approximate counterpart is what makes spectral clustering useful: if a graph is connected but has a small number of edges bridging otherwise densely connected groups of vertices, the smallest nonzero eigenvalues of L are close to zero, and their eigenvectors are close to piecewise-constant on the natural clusters, even though no exact disconnection has occurred.

Definition 5.5. Spectral clustering into k clusters proceeds as follows: compute the graph Laplacian L ; compute the eigenvectors $v_1, \dots, v_k$ corresponding to its k smallest eigenvalues; form the $n \times k$ matrix U with these eigenvectors as columns, so that row i of U is a k -dimensional embedding of vertex i ; then apply the k -means algorithm – which partitions the rows of U into k clusters by minimising within-cluster variance – to obtain the final cluster assignment.

By Theorem 5.4, if G has exactly k connected components, the k -dimensional zero-eigenspace of L is spanned by the k components' indicator vectors. For *any* choice of orthonormal basis $v_1, \dots, v_k$ of this eigenspace (which is all a general-purpose eigendecomposition routine guarantees – the returned basis need not itself be the indicator vectors, but some orthonormal linear combination of them), two vertices in the same connected component receive identical rows in U , while vertices in different components receive distinct rows; consequently the n rows of U take exactly k distinct values, one per component, and k -means on U recovers the components perfectly. In the more realistic case where the clusters are only approximately separated, this same embedding concentrates vertices from the same cluster near one another in $\mathbb{R}^k$, which is exactly the structure k -means is suited to separate.

5.1 The normalized Laplacian

Remark 5.6. In practice, spectral clustering is more commonly applied to the **normalised Laplacian** $L_{\text{sym}} = I - D^{-1/2}AD^{-1/2}$, or the closely related **random-walk Laplacian** $L_{\text{rw}} = I - D^{-1}A$, rather than the unnormalised L used above (both defined assuming every vertex has positive degree; isolated vertices, with $\deg(v) = 0$, require a separate convention, typically excluding them or adopting the conventional extension $D_{ii}^{-1} = D_{ii}^{-1/2} = 0$ when $D_{ii} = 0$). Normalisation corrects a tendency of the unnormalised Laplacian to favour splitting off small, low-degree components; the correspondence between zero eigenvalues and connected components (Theorem 5.4) carries over to L_{sym} and L_{rw} with only minor changes to the argument. Shi and Malik (2000) motivate normalised spectral clustering directly from the *normalised cut* graph-partitioning objective, and von Luxburg (2007) gives a complete tutorial derivation connecting the normalised cut relaxation to the eigenvectors of L_{sym} .

Remark 5.7. `import numpy as np`
`from scipy.sparse.csgraph import laplacian`
`from sklearn.cluster import KMeans`

```
L = laplacian(A, normed=True)           # A: adjacency matrix
eigvals, eigvecs = np.linalg.eigh(L)
U = eigvecs[:, :k]                     # k smallest eigenvalues
labels = KMeans(n_clusters=k).fit_predict(U)
```

Equivalently, `sklearn.cluster.SpectralClustering` performs the same steps (Laplacian construction, eigendecomposition, k -means on the embedding) as a single estimator with a `fit_predict` interface consistent with `scikit-learn`'s other clustering estimators.

6 Exercises

Exercise 6.1. Let $V = \{1, 2, 3, 4, 5\}$ with edges $\{1, 2\}, \{1, 3\}, \{2, 3\}, \{4, 5\}$. Compute the degree of every vertex and verify the handshake lemma directly against the number of edges. Identify the connected components of the graph, and state the dimension of the null space of the graph Laplacian that Theorem 5.4 predicts.

Solution. The degrees are $\deg(1) = 2$, $\deg(2) = 2$, $\deg(3) = 2$, $\deg(4) = 1$, $\deg(5) = 1$, summing to 8. There are 4 edges, and $2 \times 4 = 8$, confirming the handshake lemma. Vertices $\{1, 2, 3\}$ are mutually reachable via the edges among them, and vertices $\{4, 5\}$ are mutually reachable via their single edge, but no edge joins $\{1, 2, 3\}$ to $\{4, 5\}$, so the graph has exactly two connected components: $\{1, 2, 3\}$ and $\{4, 5\}$. By Theorem 5.4, the null space of the Laplacian therefore has dimension 2, spanned by the indicator vectors of these two components. $\square$

Exercise 6.2. Recompute the PageRank vector of Example 4.3 using the damping factor $\alpha = 0.85$ instead of $\alpha = \frac{1}{2}$, and compare the resulting ranking of A , B , and C to the $\alpha = \frac{1}{2}$ result. Explain, in terms of the random-surfer interpretation, why increasing α toward 1 makes the ranking more sensitive to the graph's link structure and less like the uniform distribution.

Solution. With $\alpha = 0.85$, the Google matrix is $G = 0.85M + 0.15J/3 = 0.85M + 0.05J$, where M and J are as in Example 4.3. Solving $Gr = r$ with $r_A + r_B + r_C = 1$ by the same method as in the example (substitute the equations into one another and normalise) gives approximately $r_A \approx 0.388$, $r_B \approx 0.215$, $r_C \approx 0.397$. The ordering $C > A > B$ is unchanged from the $\alpha = \frac{1}{2}$ case, but the gap between the highest- and lowest-ranked vertices has widened relative to the uniform ranking $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. This is because a larger α means the random surfer follows an outgoing link, rather than teleporting to a uniformly random vertex, with higher probability at every step; since teleportation is the only mechanism pulling the stationary distribution toward uniform, weighting it less heavily lets the graph's own link structure dominate the long-run behaviour of the surfer, and hence the ranking. $\square$

Exercise 6.3. Let G be the disjoint union of two triangles: $V = \{1, 2, 3, 4, 5, 6\}$ with edges $\{1, 2\}, \{2, 3\}, \{1, 3\}$ forming one triangle and $\{4, 5\}, \{5, 6\}, \{4, 6\}$ forming the other, with no edges between the two triangles. Write down the graph Laplacian L , and verify directly (by exhibiting two linearly independent vectors x with $Lx = 0$) that the null space has dimension 2, consistent with Theorem 5.4.

Solution. Every vertex has degree 2 (each triangle is 2-regular), so $D = 2I$. The Laplacian is

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ -1 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

Take $x^{(1)} = (1, 1, 1, 0, 0, 0)^\top$, the indicator of the first triangle. Each of the first three rows of L sums its diagonal entry (2) with its two off-diagonal entries within the block (-1 and -1), giving $2 - 1 - 1 = 0$ for each of the first three coordinates of $Lx^{(1)}$, and the last three coordinates of $Lx^{(1)}$ are zero since $x^{(1)}$ is zero on those columns and the corresponding block of L has no cross-terms. So $Lx^{(1)} = 0$. Identically, $x^{(2)} = (0, 0, 0, 1, 1, 1)^\top$, the indicator of the second triangle, satisfies $Lx^{(2)} = 0$ by the same argument applied to the second block. Since $x^{(1)}$ and $x^{(2)}$ have disjoint supports, they are linearly independent, exhibiting a two-dimensional subspace of the null space; Theorem 5.4 confirms this is exactly the null space, since G has exactly two connected components. $\square$

Exercise 6.4. Suppose spectral clustering with $k = 2$ is applied to a connected graph consisting of two dense clusters of vertices joined by a single bridging edge. Explain, using Theorem 5.4 and the discussion immediately following it, why the second-smallest eigenvalue of the Laplacian is expected to be small but strictly positive, and why the eigenvector belonging to it is expected to be approximately, but not exactly, constant on each of the two clusters.

Solution. If the single bridging edge were removed, the graph would split into exactly two connected components, and Theorem 5.4 would give a null space of dimension 2, spanned by the two components' indicator vectors: the smallest eigenvalue 0 (with multiplicity 2) would include an eigenvector distinguishing the two clusters. With the bridging edge present, the graph is connected, so by Theorem 5.4 the eigenvalue 0 now has multiplicity exactly 1 (the all-ones vector). The second-smallest eigenvalue is no longer exactly zero, but since only a single edge differs from the disconnected case, this suggests why the quadratic form $x^\top Lx = \sum_{\{i,j\} \in E} (x_i - x_j)^2$ from Proposition 5.2 should remain small for a near-constant vector x close to the disconnected case's cluster-indicator eigenvector: such an x picks up only one additional nonzero term $(x_i - x_j)^2$, from the bridging edge itself, relative to the disconnected case where it picked up none. This is intuitive rather than a rigorous bound, but it explains why the second-smallest eigenvalue is expected to remain small, and its eigenvector to remain close to, but not exactly, constant on each cluster. $\square$

Further Reading

Newman (2010) is the standard graduate-level textbook covering the whole of this lecture – adjacency structure, centrality, and spectral methods – in far greater depth, and is the natural next stop for anyone wanting to go beyond the material here. Brin and Page (1998) is the original PageRank paper, written for a general computer science audience and readable directly. Freeman (1977) is the original source for betweenness centrality as a formal, comparable measure; Freeman (1978/79) gives a widely-cited conceptual clarification unifying degree, closeness, and betweenness centrality under a common framework, though closeness centrality itself traces back earlier, to Bavelas (1950) and Sabidussi (1966). Shi and Malik (2000) introduce the normalised cut objective behind normalised spectral clustering, and von Luxburg (2007) gives a self-contained tutorial derivation of spectral clustering from first principles, including the normalised variants only briefly mentioned in §5.1 above.

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